Name _____ Period _____ Date _____

Phenomena: The Siamese Cat Mystery

Focus question: How can a difference in DNA change which parts of a cat are dark?

I notice…	I wonder…

Part 1: My initial model

Draw what is inside a Siamese cat's cells that leads to dark ears, paws, and tail.

Part 2: Key ideas

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Phenomena: The Siamese Cat Mystery (continued)

Part 3: Temperature and fur color

Body part	Skin temp (°C)	Fur color
Back		
Belly		
Upper leg		
Ear		
Nose and face		
Paw		
Tail tip		

Pattern: dark fur goes with _____ skin.

Part 4: Litters of kittens

Parents	What kittens did they have?
Siamese × Siamese	
Siamese × solid	
Solid × solid	

Exit ticket

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What Is DNA Made Of?

Focus question: What is DNA built from, and what is the message?

Warm-up

Sketch what you think DNA looks like.

Part 1: One nucleotide

Draw and label the sugar, phosphate, and base.

Part 2: Words to know

Word	In my words
double helix	
nucleotide	
backbone	
base pair	

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What Is DNA Made Of? (continued)

Part 3: Partner strands

Strand	Partner strand
A T G C C T	
T T T G A C	
C T G G G A	

The order of the bases is the _____.

Part 4: One letter apart

Solid cat: A T G G G A C T G T T T
Siamese cat: A T G A G A C T G T T T

How many letters differ? Where?

Exit ticket

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Build, Copy, and Read a Gene

Focus question: How does a gene's message become a chain of amino acids?

Warm-up

In my own words, what does “copy, then read” mean?

Part 1: Lab record

Step	What I built (letters or colors)
A: Template strand	
B: Partner strand	
C: The copy	
C: Amino acid chain	

Which strand carries the “coding” letters? How do you know?

Name _____ Period _____ Date _____

Build, Copy, and Read a Gene (continued)

Part 2: Change one letter

Step	What I found
Changed template	
New copy	
New amino acid chain	

How many amino acids changed? Which one?

Part 3: Connect to the cat

What did one letter do to the chain? Name two ways the gummy model is different from real DNA.

Exit ticket

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What Can One Letter Do?

Focus question: What can a one-letter change in DNA do?

Warm-up

Part 1: Card sort, glue each card under its category

1: Changes an amino acid in the chain	2: Changes whether or when the protein is made	3: No noticeable change

Part 2: My reasons

Card	Category	My reason
A		
B		
C		
D		
E		
F		

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What Can One Letter Do? (continued)

Part 3: The Siamese cat

DNA	Amino acid	Protein	Fur

How does this explain the dark points and the white newborn?

Claim check: “Every change in DNA changes how a cell works.” Use the cards to respond.

Exit ticket

Name _____ Period _____ Date _____

Same DNA, Different Jobs

Focus question: If all cells have the same DNA, why do cells have different jobs?

Warm-up

Part 1: Two kinds of cells

Cell	Genes in cell	Proteins made	Job
P	1, 2, 3, 4	1 and 2	Carries signals
Q	1, 2, 3, 4	3 and 4	Squeezes to move the body

Same DNA? Why different jobs?

Part 2: Cells in a Siamese cat's ear

Cell type	Has the gene?	Makes protein X?	Job
Pigment-making cell			Makes dark color
Skin cell			Covers and protects
Nerve cell			Carries signals

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Same DNA, Different Jobs (continued)

Part 3: Where does the changed protein work?

Works only in skin cooler than about 33 °C (a simplified number).

Body part	Skin temp (°C)	Protein works?	Fur color
Ear	28		
Nose	30		
Paw	28		
Tail tip	27		
Upper leg	34		
Back	36		
Belly	36		

Explain it with the words DNA, gene, protein, and job.

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Case Data

Focus question: How does a DNA change lead to an effect on the body?

Warm-up

The chain for the cat: one-letter DNA change → _____ → _____ → _____

Part 1: Case 1, Siamese and solid cats

	Solid cat	Siamese cat
Gene		
Amino acid chain		
Where the protein works		
Fur color		

Part 2: Case 2, a human gene

Person	Gene	Protein	Symptoms
1			
2			

The chain for Case 2: _____ → _____ → _____ → _____

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Case Data (continued)

Part 3: Compare the cases

How they are alike	How they are different

Part 4: Claim, evidence, reasoning

Claim: what caused the Siamese cat's dark points?

Evidence: at least two numbers from the data.

Reasoning: how does a one-letter DNA change lead to dark color only on the cooler body parts?

Exit ticket

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Review and Back to the Cat

Focus question: How can I explain the Siamese cat with DNA, genes, and proteins?

Warm-up: My CFA scores

CFA	My score	Review done?
LS1-1.1		
LS1-1.2		
LS1-1.3		
LS1-1.4		
LS1-1.5		

Part 1: Vocabulary

Word	In my words
DNA	
nucleotide	
gene	
base sequence	
amino acid	
protein	

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Review and Back to the Cat (continued)

Part 2: My revised model

Show the gene, the one-letter change, the copy-then-read path, the amino acid chain, the protein, and the dark points.

Part 3: Write It, two Siamese kittens

Why are the cool-room kitten's ears darker?

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Unit Assessment and Reflection

Focus question: What do I know about how DNA leads to traits?

Warm-up: Before the test

Three things I am sure about:

Part 1: Back to the cat

What I would add to my Day 1 model now.

Part 2: Looking back at my CFAs

CFA	My score	What helped me
LS1-1.1		
LS1-1.2		
LS1-1.3		

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Unit Assessment and Reflection (continued)

CFA	My score	What helped me
LS1-1.4		
LS1-1.5		

Part 3: What I will do next time

Exit ticket

One question I still have about DNA and proteins: